



Serveron[®] TM View[™] Standard Installation Manual

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Getting Started

Scope

This manual describes the installation procedures for TM View Standard.

TM View Standard is freely distributed with all Serveron Transformer Gas Monitors (TMx).

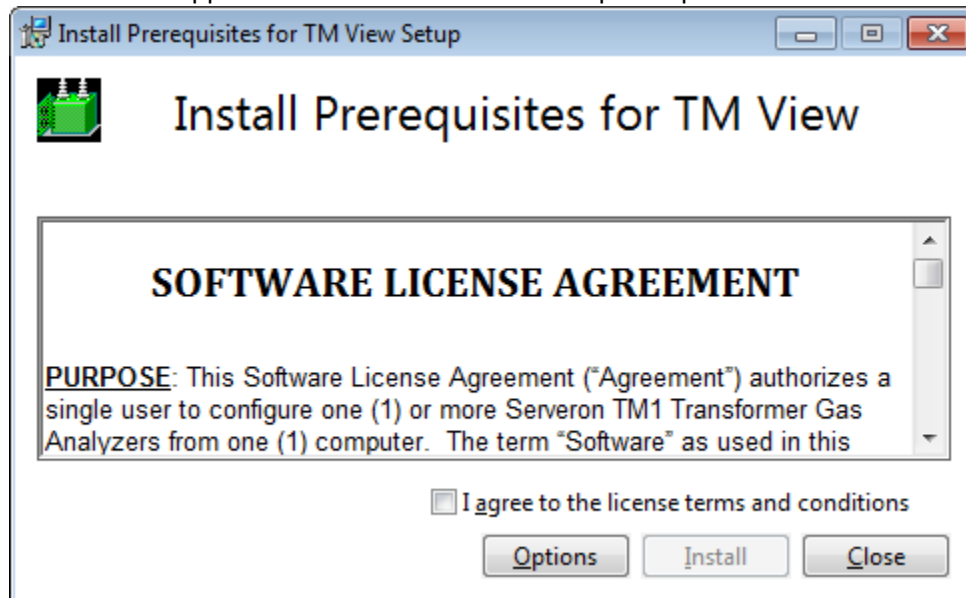
Selecting a configuration

The installer for TM View Standard installs SQL Server Express and TM View on the same computer.

1. Insert the installation CD. If your computer allows auto-run on insertion, the installer will start automatically.
If your computer does not allow auto-run of removable media, you will have to navigate to the CD and double-click on the file "Setup.bat"

*** Note: The "Setup.bat" file needs to be run only during the initial installation of TM View. If you are reinstalling TM View double click the file "Launch After Reboot of Setup.bat".

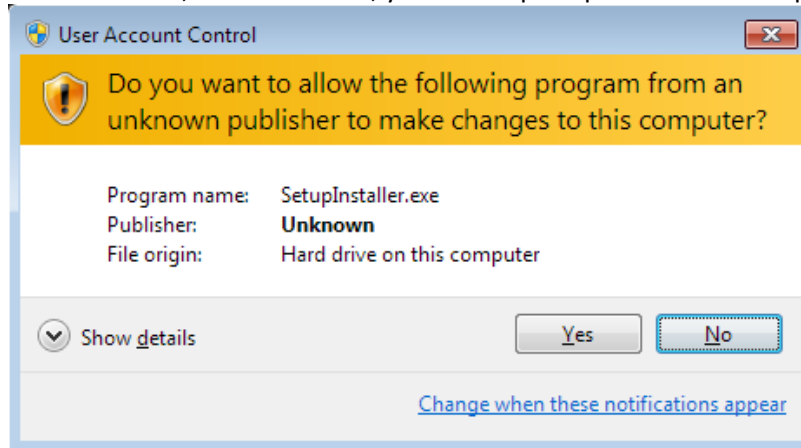
2. It launches an application which will install all the pre-requisites for TM View.



3. The pre-requisites include the .NET framework 3.5 SP1 and the .NET framework 4.0.
4. Check the "I agree..." checkbox to accept the license agreement and continue the installation.



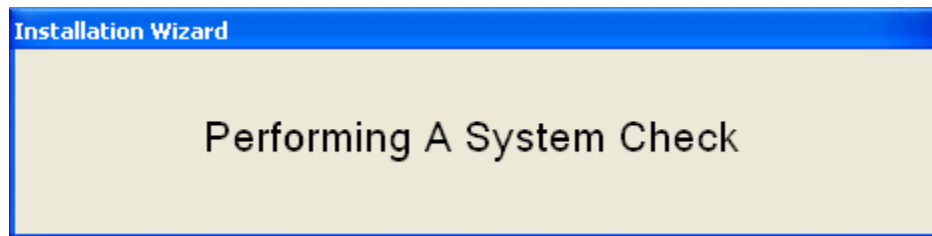
5. On Windows 7, if UAC is active, you will be prompted to allow the program to run:



6. The installer will install Microsoft .NET framework 3.5 (on Windows XP) and 4.0 (on Windows 7 and Windows XP). If the framework 3.5 is installed now, the machine will reboot after that installation, and then the installation of the framework 4.0 will be carried out.

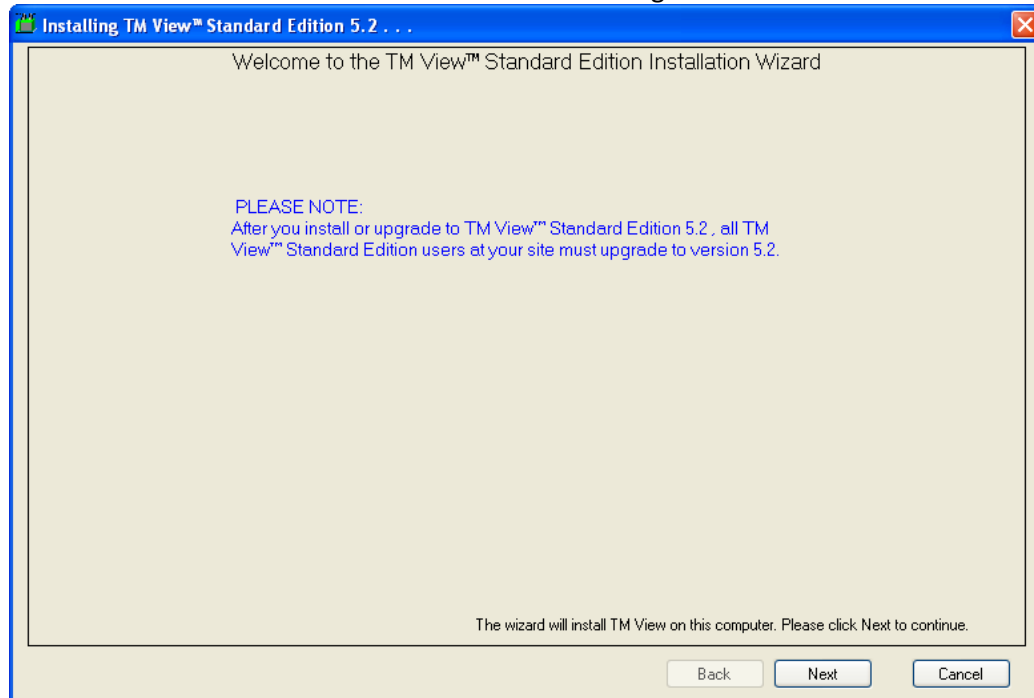
*** NOTE : If the system requires a reboot, and auto-run is active, the installer will attempt to reinstall the prerequisites. Cancel the reinstall and navigate to the CD and double click the file "Launch After Reboot of Setup.bat".

7. The prerequisites installer is complete and will automatically launch the Installation Wizard.
8. The Installation Wizard will launch a window indicating that it is doing system specification checks. These processes can take a couple of minutes to perform.

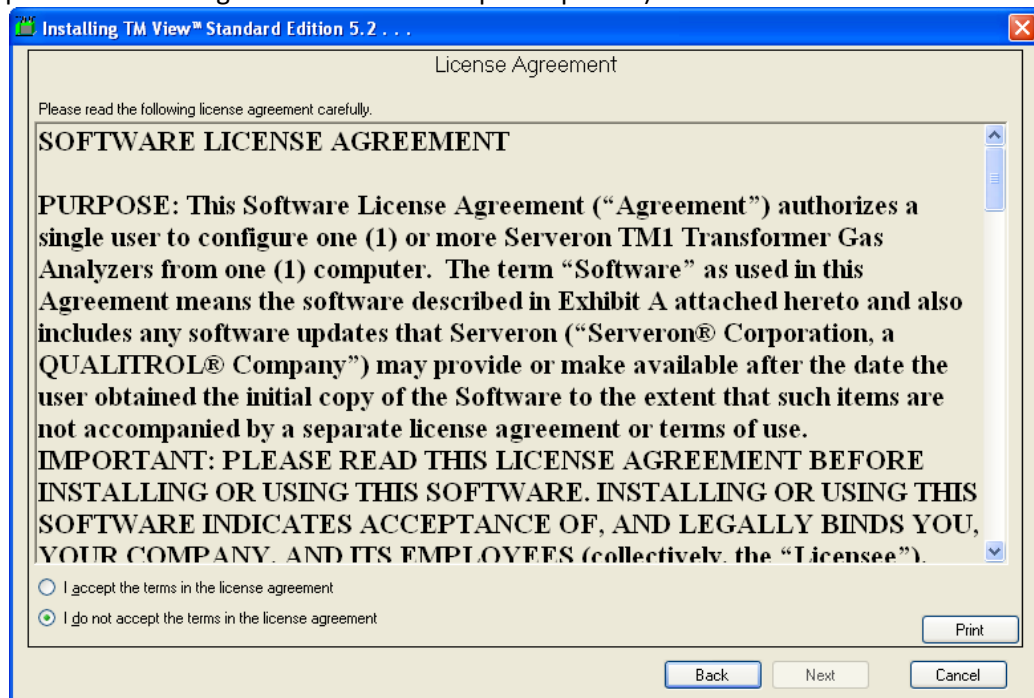




9. The installation wizard will start and show the following screen:

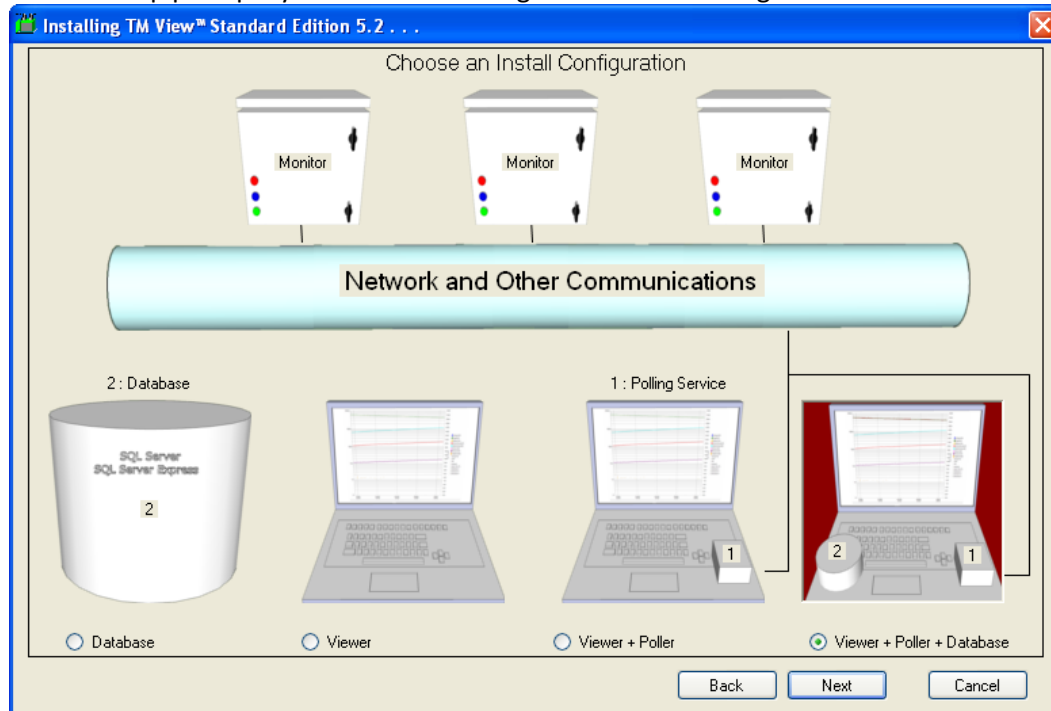


10. Click Next and you will be prompted to accept the license agreement with Serveron/Qualitrol (the previous license agreement was for the pre-requisites)





11. The next step prompts you to select a configuration for installing TM View:



12. The options are as follows, and the detailed description is found in the sections below.

- a. Database.
This option installs only the database on the current machine. Use this option if you want to install the software on another machine as the one that runs the database. This installation type is [described here](#).
- b. Viewer.
This option installs a view-only instance of TM View. Use this option if you want to install a simple viewer, without polling capabilities. This assumes that you have installed the database on another machine, and that you have or will install a poller on another machine. This is called a [“Viewer-Only” installation](#).
- c. Viewer+Poller.
This option installs the poller and the viewer for TM View. Use this if you have installed the database on another machine already, and wish to use it for this installation of TM View. This is called a [“Server” installation](#).
- d. Viewer + Poller + Database.
This option installs everything on the same machine. Use this if you want to have an entirely self-contained instance of TMN View (for example, with a laptop that you intend to use to connect to transformers in a substation. This is called a [“single user installation”](#)

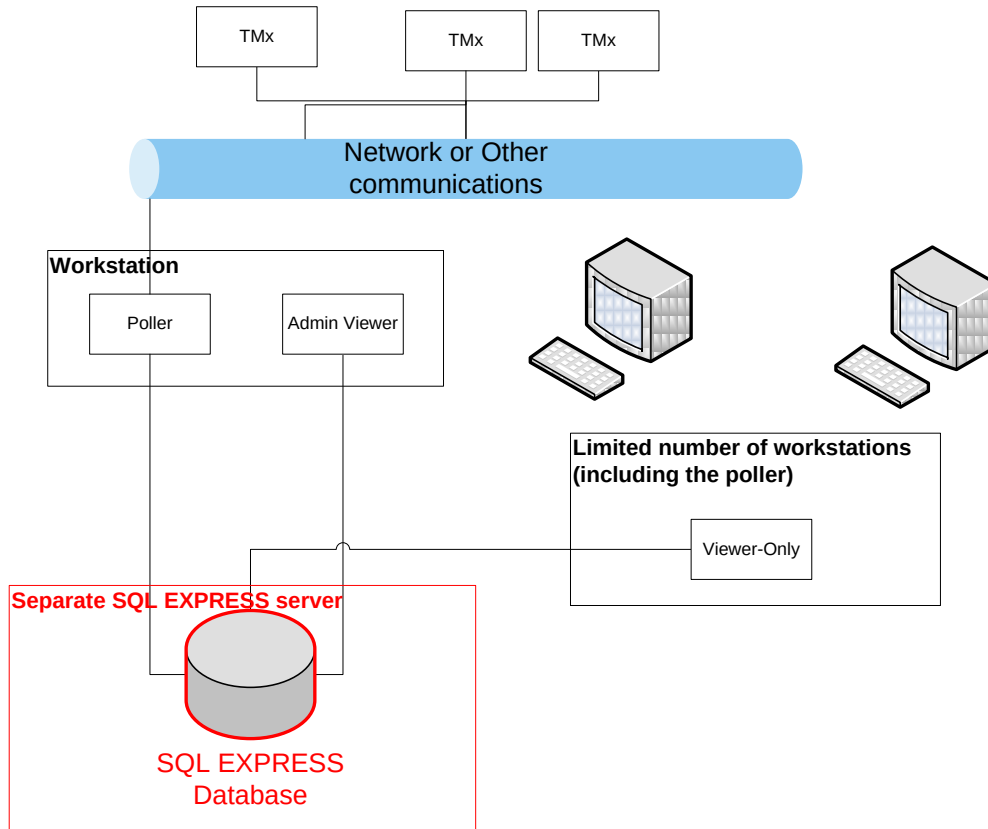
Depending on which configuration you selected, you will find yourself in one of the scenarios described below.

13. Click the picture or button on the type of installation to be installed.



14. Click the Next button.
15. The screen will change to the selected configuration option.
16. Proceed scanning this guide until the selected configuration option is reached.

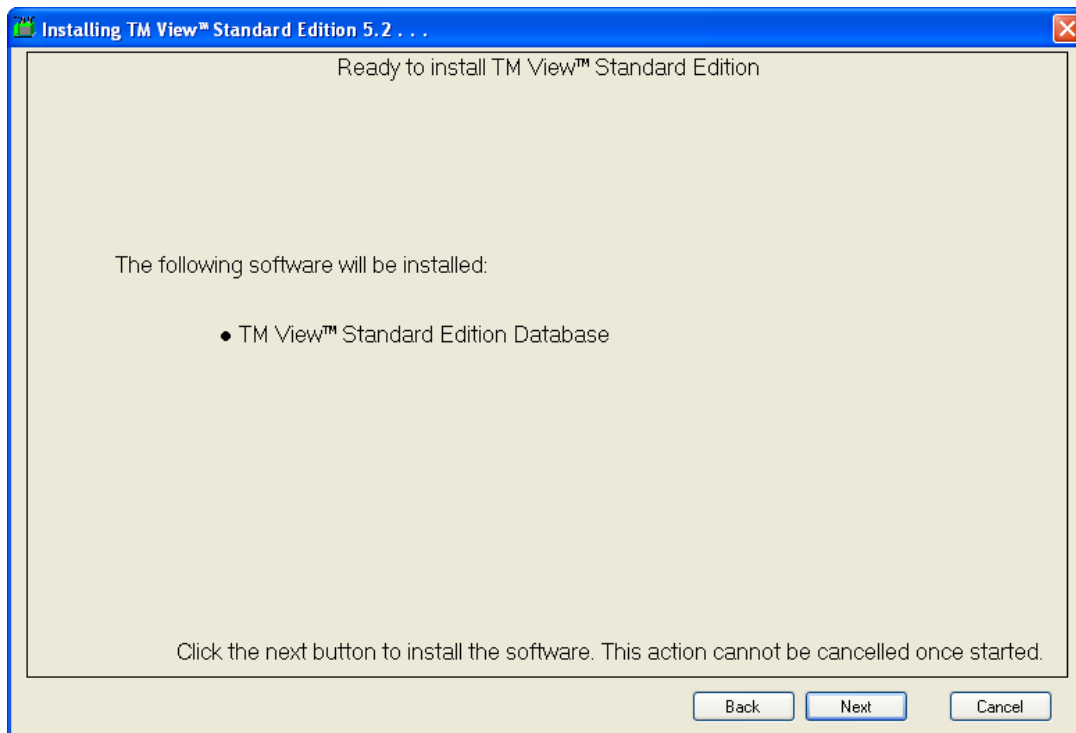
Database Only



This type of installation will create the database for TM View on the machine you run it on. If you run the installer on a machine which has already an instance of SQL Server installed, the installation will use that instance of SQL Server. It will not install another one.

By default, when the installer for TM View installs SQL Server, it creates an instance of SQL server EXPRESS called "TMVIEWSQL". If you already have an instance of SQL Server installed with another name, The installer will still create an instance called "TMVIEWSQL".

Once SQL Server is installed and the TMVIEW SQL instance created, the installer will create the database for TM View and populate it with the minimum configuration data required for TM View.

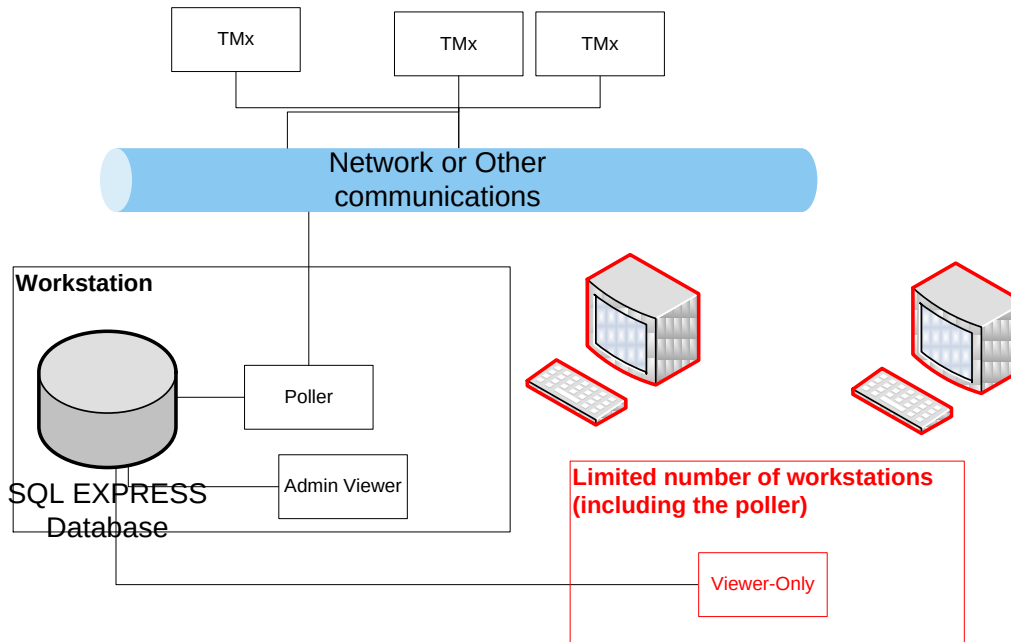


1. Click next.
2. The system will begin installing any required prerequisites needed for SQL Server Express and reboot the system.
3. After the reboot SQL scripts will be applied by the Installation Wizard to create the database.
4. The installation will complete.

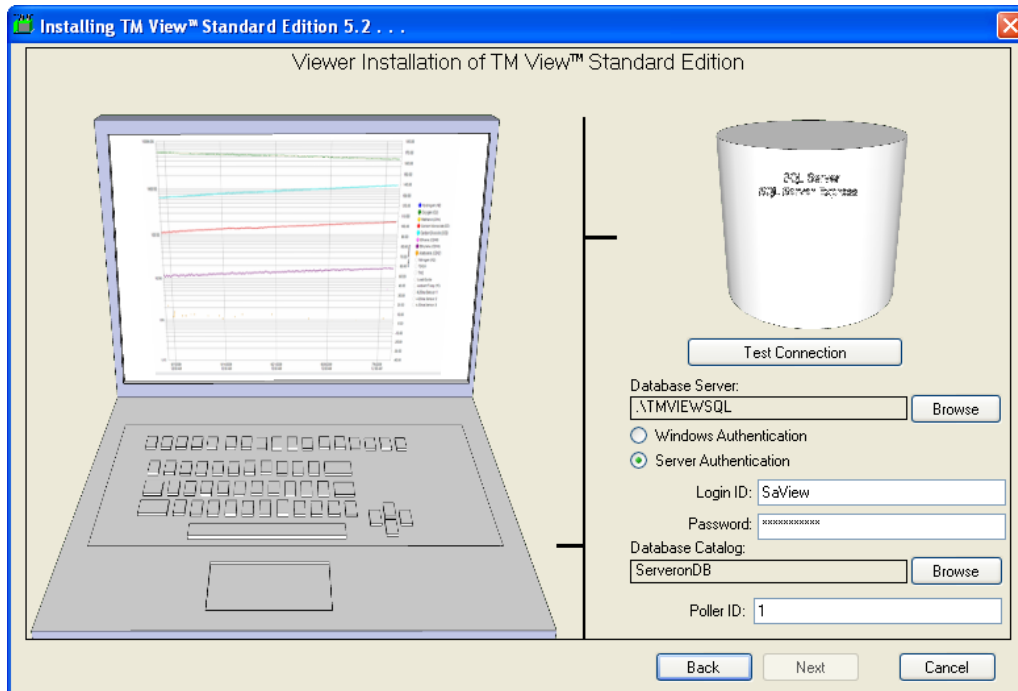


Viewer Only

The configuration will be as follows, in red outline:



This installation configuration will install TM View without a polling service or database. By default the installation is looking for a database installed on the machine.



1. Update the Database Server so that it displays the correct computer.
 - a. Click Browse
 - b. Select the Server to use.
 - c. Click OK.

*** NOTE: A more detailed explanation of this process is in the back of this Guide under Changing the Database Server.

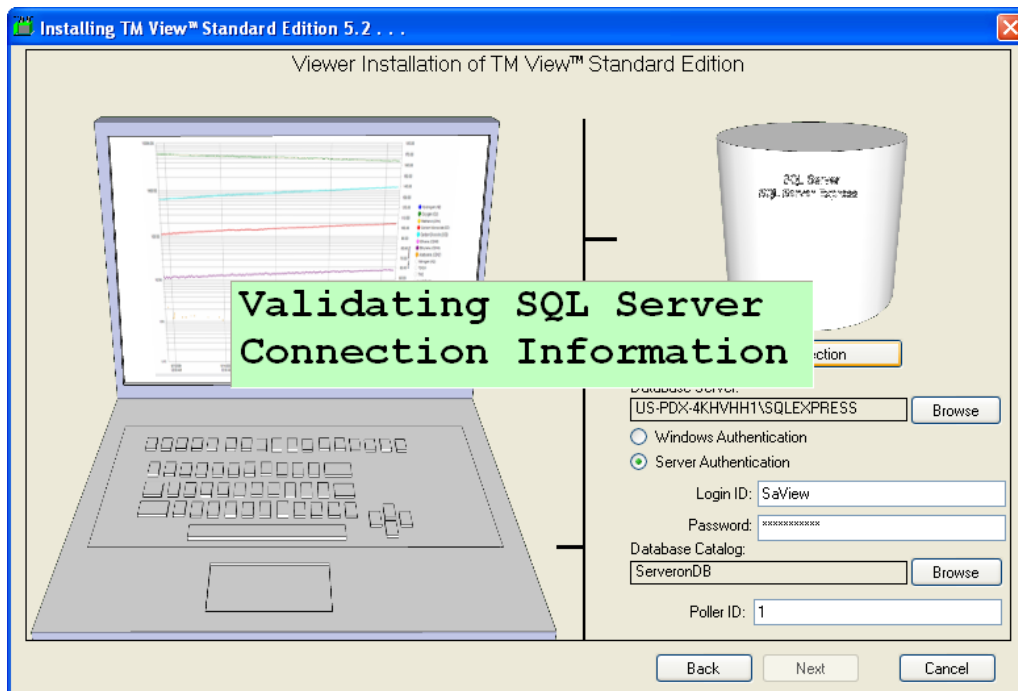
2. Choose between Windows Authentication and Server Authentication.

*** Note: Unless the database installation has been customized use Server Authentication and the supplied defaults.

3. Update the Database Catalog so that it knows the destination database.

*** NOTE: A more detailed explanation of this process is in the back of this Guide under Changing the Database Catalog.

4. Click Test connection.

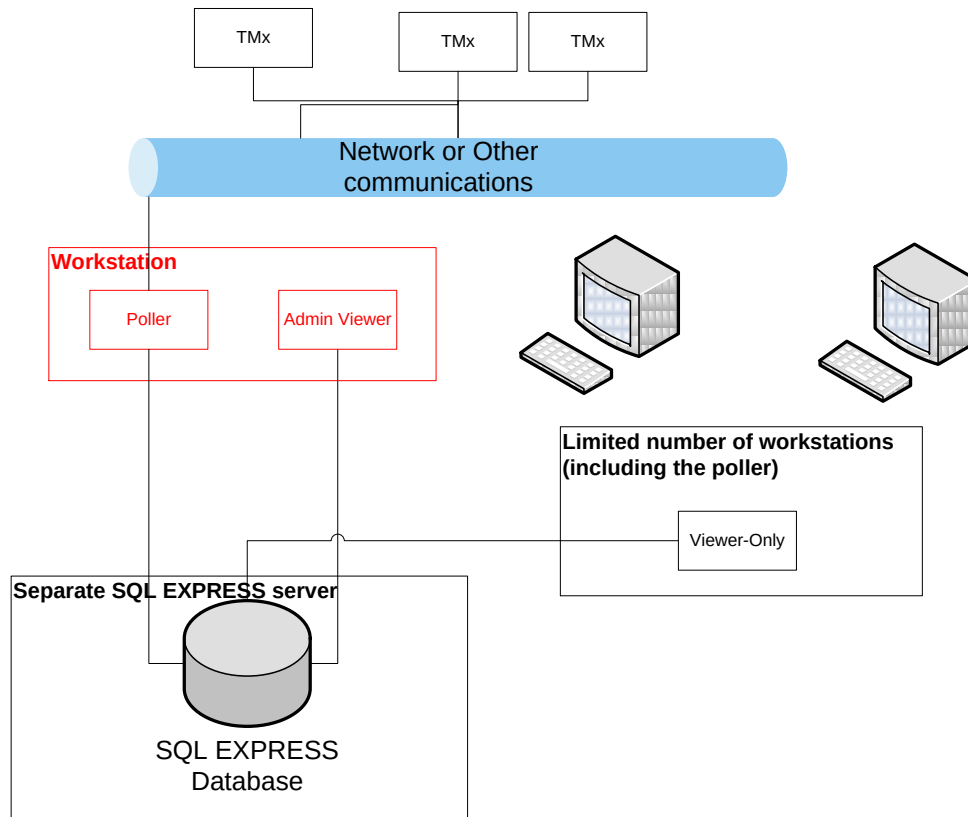


5. The connection will then display a green check mark on the database (cylinder) symbol indicated that it succeeded.

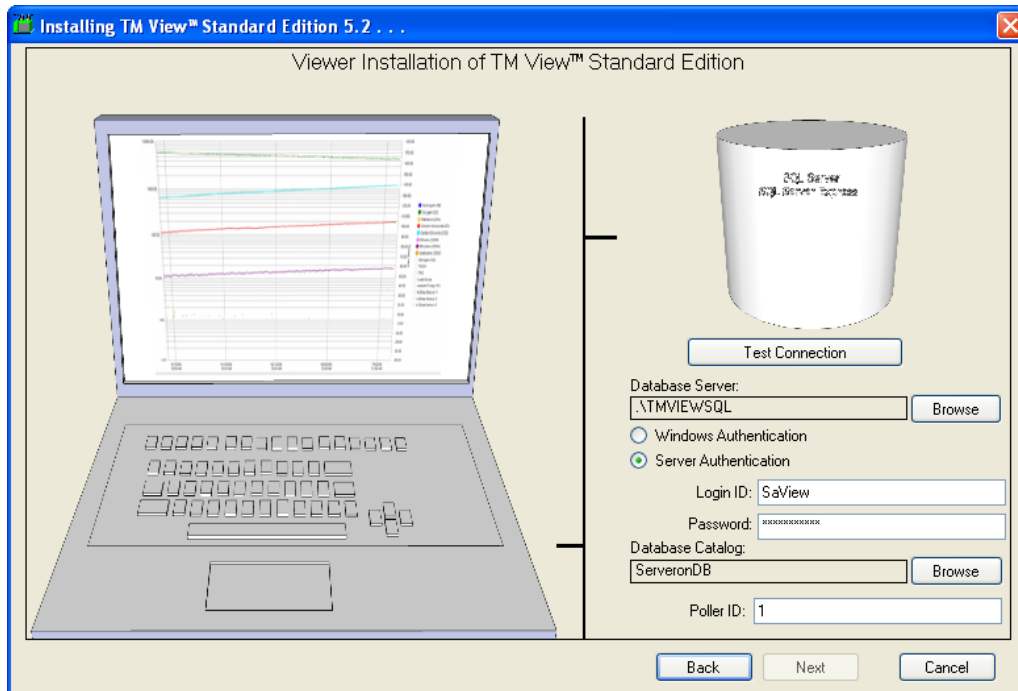
*** NOTE: If the connection fails a message will be displayed attempting to display the reason for the failure.



Server: Viewer + Poller



This installation configuration will install TM View with a Polling service. By default the installation is looking for a database installed on the machine.



1. Update the Database Server so that it displays the correct computer.
 - a. Click Browse
 - b. Select the Server to use.
 - c. Click OK.

*** NOTE: A more detailed explanation of this process is in the back of this Guide under Changing the Database Server.

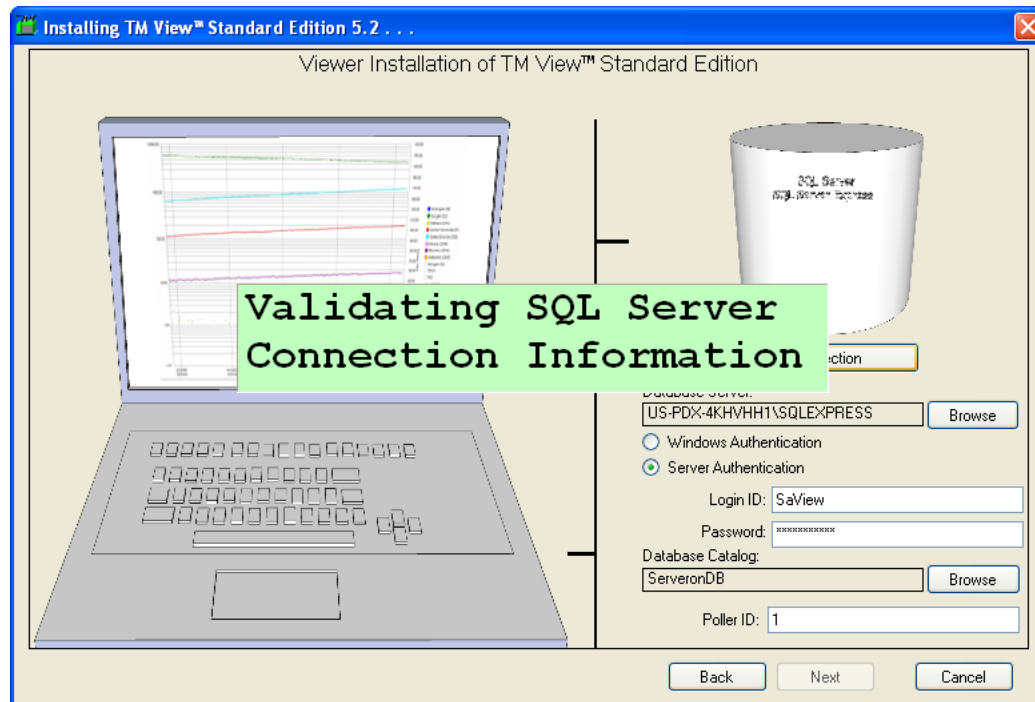
2. Choose between Windows Authentication and Server Authentication.

*** Note: Unless the database installation has been customized use Server Authentication and the supplied defaults.

3. Update the Database Catalog so that it knows the destination database.

*** NOTE: A more detailed explanation of this process is in the back of this Guide under Changing the Database Catalog.

4. Click Test connection.



1. The connection will then display a green check mark on the database (cylinder) symbol indicated that it succeeded.

*** NOTE: If the connection fails a message will be displayed attempting to display the reason for the failure.



Single User (Database, Viewer, and Poller)

The installation configuration will be as follows:

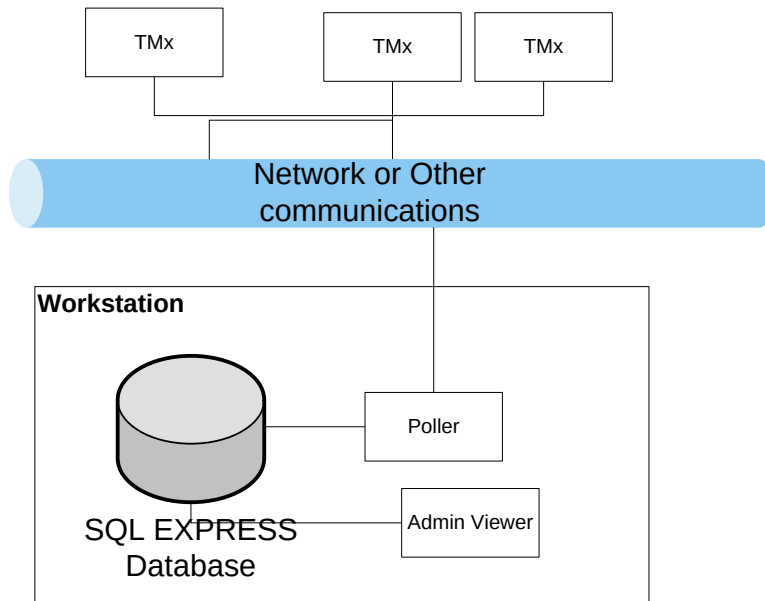
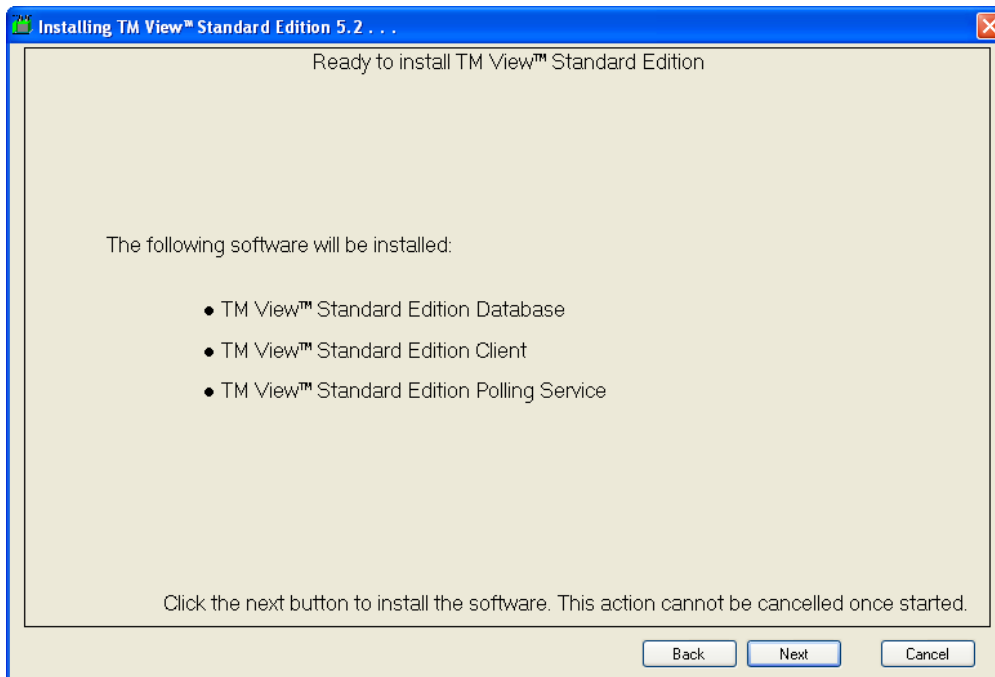


Figure 1 - Single User Installation

This procedure will install the database and all the software on the same machine.





Clicking next on this screen will install all components of TM View and any required prerequisites. If there is no database on the workstation one will be created.

*** NOTE: A reboot will be required if there is no database installed.

If a database is found then it will be upgraded to the latest database version. Finally TM View will be installed and the installation will be completed.



System Requirements

Minimum Requirements

The computer system for the single-user installation using SQL Server Express should have the minimum following specifications:

- 10 gigabytes of hard drive space
- 1.4 GHz processor
- At least 1 GB of RAM, 2 GB or more are recommended to run SQL Server Express efficiently.
- One of the following operating systems.
 - Windows Server 2003 R2 SP2
 - Windows Server 2008 SP2
 - Windows Server 2008 R2
 - Windows 7
 - Windows XP is acceptable but not recommended to run a database server.
- .NET Framework 3.5 SP1 - <http://go.microsoft.com/fwlink/?LinkId=120550>
This framework is built-in the Windows 7 operating system. If you are running Windows 7, this installation will not occur. However, you should run Windows Update to make sure you have the latest patches and security updates.
- .NET Framework 4.0. This is needed to run SQL Server and the installer wizard. It is installed with the pre-requisites.
- Windows Installer 4.5 - <http://go.microsoft.com/fwlink/?LinkId=123422>. The installer for TM View installs this.
- If you are using SQL Server 2005 or 2008 instead of SQL Server Express 2008, different system requirements are indicated, including
 - Power Shell 1.0 - <http://go.microsoft.com/fwlink/?LinkId=120552>

Best Performance Requirements

For best performance using TM View, and to ensure a responsive and pleasing user experience, Serveron recommends the following hardware requirements:

Database Server

If the database server is a dedicated machine, then it should be as close as possible to the kind of data servers sold by server manufacturers. Otherwise, it should satisfy the requirements below.

- Processor: 2 GHz or more, multiple core (dual at least, quad recommended)
- RAM: 8 GB recommended – never less than 4 GB.



- Disk space: 2.0 GB during installation, 1.4 GB after installation, plus enough room for the data TM View will grow the database by approximately 2.5 MB per monitor, per year. For example, with 100 monitors, the DB will reach 1.3 GB in 5 years.
- Software:
 - o Windows XP (SP3), Windows Server 2003 (SP2), Windows 7, Windows Server 2008. All platforms can be 32 bits or x64. Server Operating systems are recommended for best performance.
 - o Windows Installer 4.5
 - o .NET Framework 3.5 SP1
- Display: Millions of colors, 1024x768 minimum resolution

Poller Server

The machine used to install the server should, optimally, be an application server as designed for this usage by server manufacturers. The hardware requirements for best performance are:

- Processor: 2 GHz or more, multiple core (dual at least, quad recommended)
- RAM: 4 GB or more.
- Disk space: 500 MB during installation, 100 MB after installation.
- Software:
 - o Windows XP (SP3), Windows Server 2003 (SP2), Windows 7, Windows Server 2008. All platforms can be 32 bits or x64. Server Operating systems are recommended for best performance.
 - o .NET Framework 3.5 SP1
- Display: Millions of colors, 1024x768 minimum resolution

Viewer-Only

The read-only viewer has low requirements for storage, but a large, high-performance display, a multi-threading processor and lots of memory will ensure a more responsive user interface.

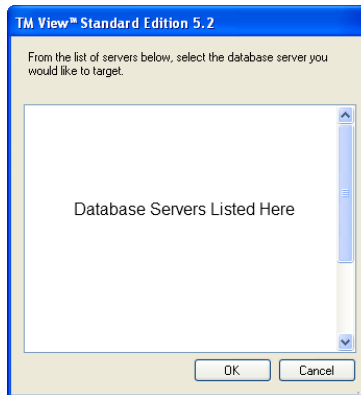
- Processor: 2 GHz or more, multiple core (dual at least, quad recommended)
- RAM: 8 GB recommended – never less than 4 GB.
- Disk space: 500 MB during installation, 100 MB after installation.
- Software:
 - o Windows XP (SP3), Windows Server 2003 (SP2), Windows 7, Windows Server 2008. All platforms can be 32 bits or x64. Server Operating systems are recommended for best performance.



- .NET Framework 3.5 SP1
- Display: Millions of colors, 1280x800 minimum resolution, 1600x1200 recommended.

Changing the Database Server:

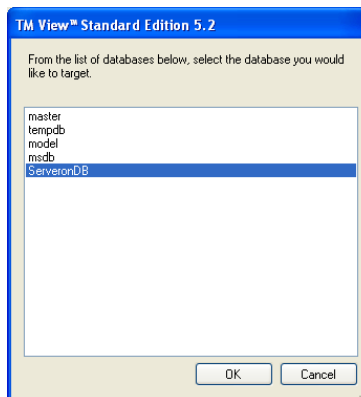
1. Click the Browse button next to the box that reads “Database Server”.
2. A browser window will appear.



3. Select a Database Server from the list of servers.
4. Click OK.
5. Click the “Test Connection” button.
6. The wizard will then attempt to validate the connection.

Changing the Database:

1. Click the Browse button next to the box that reads “Database Catalog”.
2. A browser window will appear.



3. Select a Database from the list of databases.
4. Click OK.
5. Click the “Test Connection” button.
6. The wizard will then attempt to validate the connection.



Advanced Database

Full Version of SQL Server

As an alternative to installing SQL Server Express 2008 R2, a full version of SQL Server 2005 or 2008 can be used for the database server. Please refer to your IT department and request the assistance of a DB Administrator for this case. Serveron does not support directly the installation of this configuration for TM View.

Manual SQL Server Express Instance Properties

1. In creating a custom SQL Server Express Instance certain permissions must be turned on so that the database is visible to TM View.
2. After creating the new SQL Server Instance you have to configure SQL Server Express and open the firewall for the SQL Server Express (inbound and outbound). You may require IT assistance to do this:

*** Note: From this point on the name of the SQL Server Express Instance will be called the “**Instance Name**”.

- a. SQL Server Express is installed with the TCP/IP protocol enabled and it enables the SQL Browser service, but this is not sufficient for remote operation.
- b. Run the SQL Server Configuration Manager (Start/All Programs/Microsoft SQL Server/Configuration Tools/SQL Server Configuration Manager) and select the SQL Server Network Configuration tree node.
- c. Select Protocols for the “Instance Name”, select the TCP/IP protocol, and right-click on Properties.
 - i. Go to the “IP Addresses” tab.
 - ii. Find the IP Address entry for the local host (127.0.0.1), and enable this IP address (SQL Express is installed with this IP address disabled).
 - iii. Click Apply/OK to dismiss the screen.
- d. Go to the “SQL Server Services” node and restart BOTH the “SQL Server (“Instance Name”)” service and the “SQL Server Browser” service.
- e. Open the firewall: you will need to add 2 programs to the list of allowed programs and features. Open the firewall control panel (Control Panel/Firewall) and add the following programs:
 - i. SQL Server:
C:\Program Files\Microsoft SQL Server\MSSQL10. “Instance Name”\MSSQL\Binn\sqlservr.exe
 - ii. SQL Browser:
C:\Program Files\Microsoft SQL Server\90\Shared\sqlbrowser.exe
3. When the Database properties are complete, take the CD and insert it in the computer which will run the Poller application. Run the setup.bat file.
4. Follow the installer prompts: Select the “Viewer + Poller” configuration and provide the name of the computer which is running the SQL Server Express, with the name of the instance



installed (if you selected all default options up to this point, that should be <computer name>\“Instance Name”). You may be prompted to reboot the computer.

5. Remove the installation CD and take it to another computer for a viewer-only installation.
6. On each Viewer-Only computer, and then double-click on the file setup.bat file.
7. When the installer prompts you, select the “Client/Server” button. When prompted, select the “Client” type of installation.
8. On the Database Server screen provide the name of the computer which is running the SQL Server Express, with the name of the instance installed (if you selected all default options up to this point, that should be <computer name>\TMVIEWSQL).

Custom database

TM View installs the database using default installer options for installation in a known state.

If a custom installation is desired, the following steps have been outlined. Please see your IT department for more information about executing these steps.

Using a different instance name

To install the SQL scripts on a different instance, it is recommended that you copy the whole “SQL Scripts” folder to a folder on your hard disk and follow the procedure described below.

1. Copy the “SQL Scripts” folder to your hard disk. Navigate to the destination folder.
2. That folder contains several folders:
 - a. V1.0 contains the scripts necessary to create the database create the SQL login used by TM View and create all the database objects required by TM View.
 - b. V1.1, V1.2, etc... contain the scripts necessary to update the database to successive version levels. You should always use all these scripts to bring the database up to the level required for the version of TM View you are installing.
3. Edit the file “CreateDB.bat” in the V1.0 folder.
 - a. It contains several commands for the “sqlcmd” utility. The first argument on these command lines is always “%COMPUTERNAME%\TMVIEWSQL”. That is the full name of the default SQL instance. You need to change that to the name of the instance you are using.

If that is a SQL Server Express installation, it will have the form <computer name>\<instance name>, where the computer name is the name of the server on which SQL Server Express has been installed, and the instance name is the name of the instance you have installed there: the default from Microsoft is SQLEXPRESS, but the TM View database installer installs an instance named TMVIEWSQL by default.

If that is a full SQL Server installation, it will have the form <computer name> (if you have installed SQL server with the default Microsoft instance name) or <computer name>\<instance name> if you have installed a special instance name.



- b. All the occurrences of the “%COMPUTERNAME%\TMVIEWSQL” must be changed to your own full name for the SQL Server instance you are using.
4. When the batch file has been edited and saved in the same folder, you can double-click on it, or open a CMD window, ‘CD’ to that folder and execute the batch file.
5. For all the V1.1, V1.2, etc... folders, edit the “Update.bat” batch file, and change all occurrences of “%COMPUTERNAME%\TMVIEWSQL” to the name of the instance you are using, then save the batch file in the same folder.
6. Double-click on the “Update.bat” file.
7. Repeat with all DB updates folders: V1.2, V1.3, etc..

Firewall settings

The TM View installer requires that the database is visible during installation. This is for verification of login information and to test connectivity. It may be necessary to grant the following files access to pass through the Windows firewall.

- ... MSSQL 10_50.TMVIEWSQL\MSSQL\Binn\sqlservr.exe
- ... 90\Shared\sqlbrowser.exe

Delete the database



Following these instructions will result in loss of ALL data in the database.

1. Navigate to the TM View Database installation folder.
 - a. 32 bit operating systems.
 - i. C:\Program Files\Serveron\TM View Database
 - b. 64 bit operating systems.
 - i. C:\Program Files (x86)\Serveron\TM View Database
2. Double click the Uninstall folder.
3. Double click DeleteDB.bat



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